HealthConnect Analytics — Part 1: Testing Readiness Report

AnalystLab Africa — Data Analytics Track | Week 7 Deliverable

PROJECT	PHASE	TRACK	STATUS
HealthConnect	Part 1 Audit	Data Analytics	Completed

A. Identify the Component to Be Tested

- **Tested Component:** HealthConnect Power BI Executive Dashboard & Healthcare Metrics Engine.
- **Intended Purpose:** Aggregate patient attendance, appointment wait times, lab test distributions, and department revenue into interactive visual reporting.
- **Requirement/Objective:** Verify cross-filtering integrity, correct erroneous DAX measures, and optimize report interactivity.
- **Connected Tracks:** Data Analytics Track, Data Science Track (Predictive No-Show Model), and Generative AI Track (HealthConnect KB).

B. Define the Testing Objective

To systematically audit, validate, and refine the accuracy, logical consistency, and overall usability of the Week 6 Power BI analytics dashboard outputs prior to final system integration. Specifically testing DAX calculations, data model relationships, slicer behaviors, and laboratory test distributions.

C. Comprehensive Testing & Corrective Log Matrix

COMPONENT	TEST SCENARIO	EXPECTED RESULT	ACTUAL RESULT	STATUS	ACTION TAKEN
DAX Measure	Total Department Count Audit	Dynamic count matching operational units (`200`).	Initial DAX formula locked at hardcoded value of `5` .	FAIL	Updated DAX formula to dynamic count: corrected to `200` . Retest: PASS .
DAX Measure	Total Appointment Metric	Reflect accurate total appointment volume across all facilities.	Calculated total appointment count correctly without duplicates.	PASS	Verified DAX expression against raw dataset transaction logs.
Data Model	Appointment & Service Relationship	Seamless cross-filtering between appointments and service tables.	Incorrect cardinalities produced incomplete service breakdowns.	FAIL	Re-architected 1-to-many relationship between `Appointment` and `Service` tables. Retest: PASS .
Visual Audit	Laboratory Test Distribution	Accurate distribution visual revealing top	Visual mapped incorrectly; highlighted	FAIL	Corrected visual field mappings. Verified that **Malaria Test** is the

COMPONENT	TEST SCENARIO	EXPECTED RESULT	ACTUAL RESULT	STATUS	ACTION TAKEN
		requested diagnostic procedures.	inaccurate test volume.		most conducted test. Retest: PASS.
Slicer Fix	Room Name Slicer Cleaning	Clean dropdown list without unassigned or blank room categories.	Blank / null entries appeared at top of Room Name slicer list.	FAIL	Filtered out null and blank values at visual/page level filters. Retest: PASS.
Slicer Fix	Service Name Slicers	Intuitive filtering across all service lines without canvas distortion.	Formatting alignment issues and redundant categories.	FAIL	Restructured and reformatted Service Name slicers for improved UX. Retest: PASS.

D. Testing Evidence & Artifacts

- **DAX Verification Logs:** Audited formulas for Total Department Count (`DISTINCTCOUNT(Department[DepartmentID])` adjusted to reflect total 200 units) and Total Appointments.
- **Data Model Schema Review:** Verified 1-to-many relationship integrity between `Appointment` and `Service` tables.
- **Slicer Cleaning Evidence:** Applied explicit visual-level filtering (`RoomName is not blank`) to eliminate orphaned dataset categories.